

Phase 5: Project Development Phase

Location of Source Code

The complete, executable source code for Smart Lender is present inside this same folder — "5. Project Development Phase" — in this repository. It includes the dataset, the exploratory analysis notebook, the model training script, the Flask backend, the frontend files, and the trained model artifacts.

Folder Structure

```
5. Project Development Phase/  
├─ data/  
│   └─ loan_approval_dataset.csv  
├─ notebook.ipynb  
├─ requirements.txt  
└─ src/  
    ├─ app.py  
    ├─ train.py  
    ├─ models/  
    │   ├─ loan_model.joblib  
    │   └─ scaler.joblib  
    ├─ static/  
    │   ├─ style.css  
    │   └─ script.js  
    └─ templates/  
        └─ index.html
```

Source Code Overview

notebook.ipynb performs exploratory data analysis on the loan approval dataset — loading and cleaning the data, generating summary statistics, visualizing feature correlations with a heatmap, and training a baseline classification model to validate the approach.

train.py contains the full model training pipeline: it loads and cleans the dataset, label-encodes categorical fields (education, self_employed), applies StandardScaler to numerical features, and then trains and compares three classifiers — Logistic Regression, Random Forest, and XGBoost.

The model with the highest test accuracy is automatically selected and serialized to `models/loan_model.joblib` along with the fitted scaler (`models/scaler.joblib`).

`app.py` is the Flask backend. It loads the saved model and scaler at startup, serves the frontend page at the root route (`"/`), and exposes a POST `/predict` endpoint that accepts the applicant's details as JSON, scales the input using the saved scaler, runs the prediction, and returns the approval status along with the approval and rejection probabilities.

The `static/` and `templates/` folders contain the frontend — an HTML page styled with CSS and driven by vanilla JavaScript, which collects form input, calls the `/predict` API asynchronously, and displays the result without reloading the page.

Deliverable Path

Repository: github.com/Srilatha-dot/Smart_Lender → "5. Project Development Phase" folder (complete working codebase).

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